



Doctor in astrophysics and software engineer, specialized in numerical simulation and performance optimization. Lead developer of simulation and data reconstruction pipelines for the QUBIC experiment, deployed on HPC clusters. Seeking a position in numerical simulation or HPC.

PROFIL

 tomlaclavere@gmail.com

 0643382043

 Permis B/A2

 [TomLaclavère](#)

[Website](#)

 [LinkedIn](#)

SKILLS

- Languages: C++ (advanced), Python (advanced), Fortran (intermediate)
- HPC: OpenMP, TBB, MPI (mpi4py), SLURM
 - ↳ familiar with: SYCL, Thrust, Kokkos, CUDA
- Performance: Profiling (perf, Malt, Maqao), Optimisation, Vectorisation
- Tools: Git, CMake, Docker, Linux, CI/CD, unit testing.
- Data Analysis: Scipy, Pandas, sklearn, PyTorch, JAX

SOFT SKILLS

- Rigor and work organisation
- Autonomy and adaptability
- Written and oral communication
- Ability to explain complex technical concepts
- Technical ownership and sense of responsibility

LANGUAGES

- French (native)
- English (professional)
- Japanese (beginner)

HOBBIES

- Sports: Boxing, badminton
- Programming, video games, motorcycling, hiking

TOM LACLAVÈRE

Doctor in Astrophysics · Software Engineer - Numerical Simulation & HPC

EXPERIENCE

2023-2026

CNRS |
Université
Paris-Cité |
APC Laboratory

PhD - QUBIC Experiment

- Lead developer of the simulation, reconstruction and data analysis pipeline ([qubicsoft](#), > 30k LoC, > 500 commits).
- Large-scale linear system solving: numerical methods & machine learning.
- Parallelisation: mpi4py (Python), OpenMP (Fortran).
- Deployment on SLURM cluster (up to 20 nodes, 100-200 GB of data processed per run).
- Development of calibration and noise analysis tools for data validation.

2025-

GitHub
Project

HPC / Simulation Project - 3DPhysicsEngine (C++23)

- 3D rigid body simulation engine: custom math library, two-phase collision pipeline, three integrators (Euler, Verlet, RK4)
- Benchmarks: 150+ configurations (solver × time step), convergence, performance and energy drift quantified
- Current phase: CPU optimization (SIMD, OpenMP); roadmap: CUDA/SYCL, MPI

EDUCATION

2023-2026

PhD in Fundamental Physics - Astrophysique

- QUBIC Data Analysis: Realistic astrophysical components reconstruction and atmospheric mitigation using spectral imaging.(APC)

2025

Gray Scott Summer School - HPC (CC-FR/LAPP)

- TBB, OpenMP, OpenMPI, SYCL, CUDA, Thrust
- C++, Fortran, Rust, Python, Julia

2023

Master Nuclei, Particles, Astroparticles & Cosmology (NPAC) - Mention bien ([Université Paris-Cité](#))

2021

Bachelor's in Fundamental Physic - Mention Très bien ([Université Paris-Cité](#))

2018-2020

CPGE, Physics-Chemistry - Admitted to Centrale ([Cité Scolaire Bertran-de-Born](#))

TEACHING

2024 - 2026

CentraleSupélec
Université
Paris-Cité
Ecole
d'Ingénieur
Denis Diderot

Adjunct Teaching Assistant

- Cosmology - 1ère année - Tutorials (35h) - 2026
- Physique Numérique - Master 1 - Tutorials (36h) - 2024 & 2025
- Champs Electromagnétique - 1ère année - Tutorials (24h) - 2024
- Bruits - 2ème année - Lab sessions (24h) - 2024